

AI S2025; Scheduled Quiz-II (20M)

Your Roll No:

17th Mar 2025

Your Name:

Section-I: Objective-type questions.

A correct answer will get +1 and the wrong answer will get a -0.25 mark. Your answers should be written on the next page. Do not write anything on this page, except your roll number and name. (10 M)

1. Which of the following problems can be modeled as a Constraint Satisfaction Problem (CSP)?
A) Sudoku
B) Map Coloring
C) Job Scheduling
D) **All of the above**
2. In a CSP, constraints define:
A) The variables that can be used
B) **The relationships between variables**
C) The number of possible solutions
D) The algorithm used to solve the problem
3. Which heuristic selects the region with the fewest possible color choices first?
A) Degree Heuristic
B) Forward Checking
C) **Minimum Remaining Values (MRV)**
D) Least Constraining Color
4. What is the key idea behind the Least Constraining Color heuristic?
A) **Choose a color that leaves the maximum number of choices for neighboring regions**

- B) Choose a color that is most frequently used
- C) Choose a color that is randomly assigned
- D) Choose a color based on alphabetical order

5. **In the Cryptarithmic problem:**

SEND + MORE = MONEY, what must be true about the leftmost letter in the solution?*

- A) It can be any digit from 0-9
- B) It must be greater than 5
- C) **It cannot be 0**
- D) It must always be an even number

6. **Which data structure is primarily used to store arcs in the AC-3 algorithm?**

- A) Stack
- B) **Queue**
- C) Linked List
- D) Binary Search Tree

7. **Simulated Annealing is inspired by which physical process?**

- A) Nuclear fission
- B) Chemical reactions
- C) **The cooling process of metals**
- D) Electrical conductivity

8. **How does Simulated Annealing compare to Genetic Algorithms in terms of optimization?**

- A) **Simulated Annealing maintains a single candidate solution, while Genetic Algorithms work with a population of solutions**
- B) Simulated Annealing always finds the best solution, while Genetic Algorithms do not
- C) Genetic Algorithms are faster than Simulated Annealing in all cases

D) Simulated Annealing requires crossover and mutation, while Genetic Algorithms do not

9. How does Alpha-Beta Pruning improve the efficiency of the Minimax algorithm?

A) By completely eliminating the need for recursion

B) By avoiding the exploration of nodes that will not affect the final decision

C) By increasing the branching factor

D) By skipping the Minimax evaluation step

10. What is the worst-case time complexity of the Minimax algorithm with Alpha-Beta Pruning, assuming no pruning occurs? (b: branch factor and d: Depth)

A) $O(b^d)$

B) $O(d^b)$

C) $O(b^{(d/2)})$

D) $O(\log b)$

[illegible]

Section-II: Fill in the blanks and no negative marks. Justify your answer in 1-2 sentences. (10 M)

1. Consider local search in continuous space. Let

$f(x) = (x^2 - x) + 1, x \in R$ be an objective function. For an initial point at $x=1$, the gradient of $f(x) = \underline{\hspace{1cm}} \textcolor{red}{1} \underline{\hspace{1cm}}$

Justification: $\text{derivative of } f(x) = 2x - 1 = 1$

2. Consider the problem of placing k knights (horses) on an n -by- n chessboard such that no two knights are attacking each other, where k is given and $k \leq n^2$. Suppose, we define a variable corresponding to each knight.

Then, the possible values of each variable are

$\text{Each variable's domain is a set of squares.}$

Justification:



3. Consider the right-side map. The number of possible solutions for the map-coloring problem are $\underline{\hspace{1cm}} \textcolor{red}{18} \underline{\hspace{1cm}}$ with three colors.

Justification:

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There are 18 solutions for coloring Australia with three colors. Start with SA, which can have any of three colors. Then moving clockwise, WA can have either of the other two colors, and everything else is strictly determined; that makes 6 possibilities for the mainland, times 3 for Tasmania yields 18.

4. Let X and Y are two variables with the same domain $D_X = D_Y = \{0, 1, \dots, 9\}$ set of digits with constraint $Y = X^2 + 1$. Is X arc-consistent with Y ?. If not, write updated $D_X =$ $\{0,1,2\}$ and $D_Y =$ D_Y such that X is arc-consistent with Y .

Justification: X is not arc-consistent with Y . So. the domain of X will change, i.e., $D_X = \{0,1,2\}$. There will be no change in D_Y .

5. Suppose X and Y are two variables with distinct values having the same domain $D = \{1, 2, 3\}$. Write the formulation of constraints on X and Y :

$$X \in \{1,2,3\}, Y \in \{1,2,3\}, _X \neq Y$$

Justification: The constraint $x \neq y$ ensures that the values assigned to x and y are distinct. If x is assigned a value, say 1, then y must be assigned one of the remaining values (in this case, 2 or 3), ensuring that x and y do not share the same value.

Example of possible assignments:

- If $x=1$, then y can be either 2 or 3.
- If $x=2$, then y can be either 1 or 3.
- If $x=3$, then y can be either 1 or 2.

Thus, there are 6 valid assignments